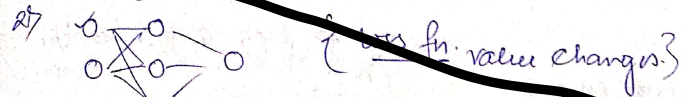


Drawbacks

1. Convergence $\rightarrow$ delay

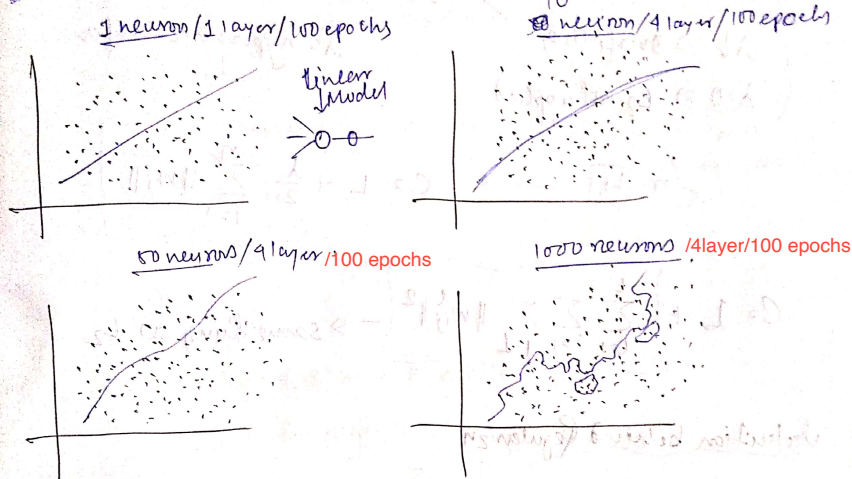


$\rightarrow$ Lf - has changed $\rightarrow$ gradient calculation complicated.

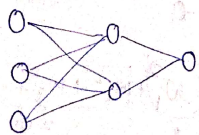
Regularization

add penalty term to loss function and discourage large weights

Why ANN model overfit?



more neurons $\rightarrow$ more complexity $\rightarrow$ capable to capture minute patterns from the dataset.



sometimes capture unnecessary patterns that are not the actual essence of the data.

Ways to solve overfitting:

Adding more data

- $\rightarrow$ Add more rows
- $\rightarrow$ Data augmentation

Reducing the complexity of the model

- $\rightarrow$ Dropout
- $\rightarrow$ Early stopping
- $\rightarrow$ Regularization $\rightarrow$ L_1 , L_2 , $L_1 + L_2$

ANN $\rightarrow$ w, b
 $\rightarrow$ minimizing loss fn

$L = \text{mse}$

$\hookrightarrow$ binary-cross-entropy

$$C = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i) + \text{penalty term}$$

[L2] $\rightarrow$

$$C = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i) + \frac{\lambda}{2n} \sum_{i=1}^n \|w_i\|^2$$

Causes

$\lambda \uparrow \Rightarrow$ underfitting $\leftarrow \left(\frac{\lambda}{2n} [w_1^2 + w_2^2 + \dots + w_n^2] \right)$
 $\lambda \downarrow \Rightarrow$ overfitting
 $\lambda = 0 \Rightarrow L_2$ not applied

Hyperparameter
 $\lambda \rightarrow 0$ to 0.01

λ hyperparameter

for [L1] $\rightarrow C = L + \left[\frac{\lambda}{2n} \sum_{i=1}^n \|w_i\| \right]$

$$C = L + \left[\sum_{l=1}^L \sum_{i=1}^n \sum_{j=1}^n \|w_{ij}^l\|^2 \right] \rightarrow \text{same thing as } L_2$$

Intuition behind Regularization

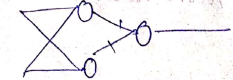
L1 $\rightarrow$ feature selection
L2 $\rightarrow$ for deep learning

$$w_1 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

$$L' = L + \frac{\lambda}{2n} \sum \|w_i\|^2$$

$$\frac{\partial L'}{\partial w_0} = \frac{\partial L}{\partial w_0} + \frac{\lambda}{2} \times 2 w_0$$

$$= \frac{\partial L}{\partial w_0} + \lambda w_0$$



$$w_1^2 + w_2^2 + \dots + w_n^2$$

differentiate w.r.t. w_1

L2 $\rightarrow$ Weight decay

$$\therefore w_1 = w_0 - \eta \left(\frac{\partial L}{\partial w_0} + \lambda w_0 \right)$$

$$= w_0 - \eta \lambda w_0 - \eta \frac{\partial L}{\partial w_0}$$

$$= (1 - \eta \lambda) w_0 - \eta \frac{\partial L}{\partial w_0}$$

smaller than previous w_0

weight decay

$\left\{ \begin{array}{l} \text{In each iteration} \\ \text{the value of } w_1 \\ \text{gradually decreases.} \end{array} \right\}$

L1 $\rightarrow$ weight becomes zero
L2 $\rightarrow$ weight becomes small not zero